

Mehmet Göktuğ Öztürk, MSc

Ecologist | Geospatial Analyst

e-mail: m.goktu66@gmail.com
website: <https://r-delisi.github.io>
github: [mehmetgoktug](https://github.com/mehmetgoktug)

orcid: [0000-0002-8507-9015](https://orcid.org/0000-0002-8507-9015)
researchgate: [Mehmet Göktuğ Öztürk](#)
linkedin: [Mehmet Göktuğ Öztürk](#)

About Me

Geospatial analyst and ecologist with five years of experience at İzmir Metropolitan Municipality, applying GIS, remote sensing, and spatial analysis to EU-funded and municipal projects (Horizon Europe CLIMAAX, CARDIMED, Interreg NEXT MED).

Advanced R programmer with skills in statistics, machine learning, and spatial/non-spatial data analysis. Comfortable across major GIS ecosystems (QGIS, ArcGIS, GRASS GIS, GEE) and experienced in reproducible workflows with R, Python, and Bash. MSc thesis on Spatiotemporal Patterns of Lightning-Induced Wildfires in Turkey (Best Master's Thesis Award, 2025). Currently a PhD student at Hacettepe University, seeking full-time geospatial roles.

Experience

Biologist - İzmir Metropolitan Municipality

March 2021 - December 2025 | İzmir, Türkiye

Contributed to EU-funded and municipal projects on climate resilience and nature-based solutions (NBS), focusing on spatial analysis, environmental assessment, and stakeholder engagement.

- **Municipal sustainability programs** (Mera İzmir, Başka Bir Tarım Mümkün, Bir Fidan Bir Dünya, Yaşayan Parklar): Performed statistical and geospatial analyses and produced biodiversity and habitat maps to inform municipal planning on pastoralism, agroecology, and afforestation.
- **CLIMAAX** (Horizon Europe, GA 101093864): Spatial data preprocessing and stakeholder coordination for İzmir's climate risk and vulnerability assessment; co-author of the official M6 deliverable (DOI: 10.5281/zenodo.15105769).
- **UCRP** (ICLEI - Zurich Insurance partnership): Spatial data preprocessing and stakeholder facilitation with local institutions and community representatives.
- **HORIZON CARDIMED**: Contributed to proposal development; field implementation and monitoring of halophytic NBS at the İzmir demo site.
- **Interreg NEXT MED**: Supported project development and reporting.

Undergraduate Research Assistant - Hacettepe University

September 2015 - January 2021 | Ankara, Türkiye

Worked on fire ecology, seed germination, and plant functional traits projects at the Functional Ecology Lab (FEL).

Field Biologist - Yolda Initiative

March 2019 - March 2020 | Mersin, Karaman, Konya, Ankara, Türkiye

Performed field studies on pastoralist goat herds in Mersin as part of the Nomadic Pastoralism Effects on Seed Dispersal Project, followed by laboratory analyses. Project run in partnership between Yolda Initiative and Hacettepe University.

Field Biologist - Doğa Derneği - Birdlife Türkiye

September 2018 - September 2019 | Adana, Türkiye

Conducted raptor migration counts at Sarımaçi station as part of the Egyptian Vulture New LIFE project (LIFE16 NAT/BG/000874).

Skills

Programming:	R (advanced), Python, Bash, Julia, SQL
GIS & Remote Sensing:	QGIS, ArcGIS, GRASS GIS, SAGA GIS, Google Earth Engine, GDAL/OGR
Other Tools:	Git, LaTeX, Office Programs
Methods:	Spatial Analysis, Remote Sensing, Machine Learning, Statistics, Ecological Modelling
Environmental Applications:	Biodiversity Assessment, Habitat Mapping, Climate Risk Assessment, Nature-based Solutions, Stakeholder Engagement
Languages:	Turkish (native), English (B2)

Certifications & Training

- OpenGeoHub Summer School 2023 “Processing and visualizing large geospatial data using R, Python and Julia”
August 2023 | Poznan, Poland
- Introduction to R Programming in the Natural Sciences Workshop in METU, **November 2018 | Ankara, Türkiye**

Education

Doctor of Philosophy - PhD, Ecology

2025 - Present | Hacettepe University | Ankara, Türkiye

Master of Science - MS, Ecology

2021 - 2024 | Hacettepe University | Ankara, Türkiye

Thesis: Spatiotemporal Patterns of Lightning-Induced Wildfires in Turkey (in Turkish with English summary).

Bachelor's degree, Biology

2015 - 2021 | Hacettepe University | Ankara, Türkiye

Volunteering

Volunteer field biologist (2015–2020) at various organizations including METU OKGT, WWF-Türkiye (EBBA2), and the Ministry of Agriculture and Forestry, contributing to bird and large mammal surveys.

Selected Publications

Öztürk, M.G., Bekar, İ., Tavşanoğlu, Ç., 2024. Rethinking lightning-induced fires: Spatial variability and implications for management policies. *Forest Ecology and Management* 572 122262. <https://doi.org/10.1016/j.foreco.2024.122262>

Öztürk, M. G., Bekar, İ., Tavşanoğlu, Ç., 2025. Modeling Wildfire Occurrence in Türkiye: The Role of Climate, Productivity, and Human Influence. In: *Proceedings of the 11th Ecology and Evolutionary Biology Symposium*, Ankara, Türkiye.

Awards

Ecology and Evolutionary Biology Association Best Master's Thesis Award [2025]

References

Prof. Dr. Çağatay Tavşanoğlu ctavsan@hacettepe.edu.tr
Hacettepe University, Türkiye

Dr. İsmail Bekar ismail.bekar@tum.de
Technical University of Munich, Germany

Dr. Hayrettin Onur Bektaş hayrettinonur.bektas@aski.gov.tr
Ankara Water and Sewerage Administration (ASKİ), Ankara, Türkiye

Engin Yılmaz engin@bican.net
Yolda Initiative, Ankara, Türkiye